

LAS⁺: Lattice-based Adaptor Signature with Stronger Security

Jiaheng Wang, Guomin Yang, Haiyang Xue, Meng Hao, and Robert Deng

Singapore Management University, Singapore
{jh.wang.2024,gmyang,haiyangxue,menghao,robertdeng}@smu.edu.sg

Abstract. Adaptor signatures (AS) enable conditional signing and atomic release of secrets, and they are widely used in payment channels, atomic swaps, oracle-based payments, and other applications. In ESORICS’20, Esgin et al. proposed LAS, the first post-quantum AS based on M-SIS and M-LWE assumptions, and demonstrated its applications to payment-channel networks and atomic swaps.

To better capture realistic attacks across diverse applications, in EURO-CRYPT’24, Gerhart et al. introduced a stronger security model formalizing four properties: *extractability*, *unique extractability*, *unlinkability*, and *pre-verify soundness*, where *extractability* encompasses unforgeability and witness extractability from earlier frameworks, and the other three are new properties required by different AS applications.

In this work, we show that although LAS achieves unforgeability and witness extractability in its original model, it lacks the newly formalized *unlinkability* and *pre-verify soundness*. Regarding unlinkability, we present a polynomial-time distinguisher based on boundary-density statistics that distinguishes standard and adapted signatures with non-negligible probability. To bridge this gap, we present LAS⁺. Recognizing that standard unlinkability is inherently difficult in the lattice setting, we first formalize a *weak unlinkability* notion, which remains sufficient to guarantee on-chain privacy for practical applications. We achieve this by modifying the signing process via a tuned rejection-sampling step. Furthermore, we attain *pre-verify soundness* by incorporating a non-interactive zero-knowledge (NIZK) proof into the underlying hard relation. Our LAS⁺ satisfies *extractability*, *unique extractability*, *weak unlinkability*, and *pre-verify soundness* under the M-SIS and M-LWE assumptions. When applied in UTXO-based atomic swaps, its restored *weak unlinkability* and *pre-verify soundness* yield stronger on-chain privacy compared to the original LAS scheme.

Keywords: Adaptor Signatures · Lattice · Stronger Security · Atomic Swap

1 Introduction

Adaptor signatures (AS), first introduced by Poelstra [13] and formalized by Aumayr et al. [1], extend digital signatures by enabling conditional signing. In

an AS scheme, a signer produces a "half-locked" *pre-signature* that cannot be completed without the correct witness to a public relation. Anyone knowing the witness can *adapt* the pre-signature into a full signature, which inherently reveals the witness to the original signer.

AS schemes have seen broad adoption in blockchain applications, including cross-chain atomic swaps [18], payment channel networks [12], coin-mixing [7,14], and oracle contracts [11]. For instance, in a fair exchange, a buyer conditionally pays using a pre-signature. To claim funds on-chain, the seller applies a secret witness to generate a valid signature. Crucially, publishing this signature automatically reveals the witness back to the buyer. This cryptographic lock-and-reveal mechanism ensures trustless atomicity without complex scripts.

Several AS based on classical hard problems, such as the discrete logarithm, have been proposed in the literature [7, 11, 14, 16–18]. However, with the advent of quantum computing, classical hardness assumptions no longer provide long-term security guarantees. This motivates the migration of classical cryptography towards Post-Quantum Cryptography (PQC), which relies on assumptions conjectured to resist quantum attacks, including those based on lattice-based, code-based, and isogeny-based problems. In particular, lattice-based schemes, such as the signature Dilithium [3], have emerged as leading PQC candidates due to their efficiency and strong worst-case hardness guarantees.

Esgin et al. [4] introduced the first post-quantum AS (denoted by LAS hereafter) based on the Module-SIS and Module-LWE assumptions, and demonstrated its application in post-quantum atomic swaps and payment channel networks. The LAS construction is an extension of a simplified version of Dilithium [3]. They also pointed out a lattice-specific obstacle in the context of AS: the zero-knowledge proofs used to certify that a secret witness is short suffer from an inherent knowledge gap, in the sense that an extracted witness only satisfies a relaxed relation R' rather than the original relation R . Since AS crucially rely on extracting a witness that is valid for the hard relation in order to adapt pre-signatures, they adjusted the AS definition and security model by working with the extended relation R' .

Meanwhile, beyond advancing post-quantum security, concurrent research efforts have also focused on strengthening the security models of AS. As illustrated by Dai et al. [2], earlier adaptor signature formulations [1] capture unforgeability and witness extractability, but their security games permit only a single pre-signing query, which cannot guarantee security in applications, such as oracle-based payments, where the adversary may acquire multiple pre-signatures on the same message. Moreover, AS generated under the early frameworks could be distinguishable from ordinary signatures. Consequently, in applications such as atomic swaps, an outsider can infer that two signatures originate from the same adaptor-based swap and link them as belonging to the same transaction. To close these gaps, Dai et al. proposed a strengthened framework that allows multiple pre-signing queries in the security game and formalized the notion of unlinkability for AS.

Building on Dai et al.’s work [2], Gerhart et al. [6] further streamlined the framework and consolidated the security notions into *unique extractability*, *unlinkability*, *extractability*, and *pre-verify soundness*. They also demonstrated the inadequacy of previous definitions by showing concrete attacks in real-world blockchain settings such as coin mixing, oracle-based payments, and blindhub protocols. For instance, the lack of *unique extractability* allows a malicious user in a coin-mixing protocol [7] to turn one pre-signature into different valid signatures and thus bias the mixing outcome. These findings show that stronger security notions are not just theoretical refinements, but necessary for preserving privacy and fairness in blockchain applications, which earlier AS schemes largely overlooked.

Motivation. While Esgin et al. [4] proposed the first post-quantum adaptor signature, LAS, based on lattices, its security was proven only in a weaker model [1]. This gap motivates our effort to revisit LAS in the stronger security model formalized in [6].

1.1 Contributions and Technical Overview

Our contributions are threefold. First, we identify security gaps in the original LAS scheme, demonstrating that it fails to achieve both *unlinkability* and *pre-verify soundness*. Second, we formalize the notion of *weak unlinkability* for the lattice setting and propose a revised scheme, LAS⁺, which resolves these issues through two core enhancements: a rejection-sampling procedure and an NIZK-augmented hard relation. Lastly, we formally prove that LAS⁺ satisfies *extractability*, *unique extractability*, *weak unlinkability*, and *pre-verify soundness* under the stronger security model.

- **Identifying Security Gaps in LAS.** We revisit the lattice-based adaptor signature scheme LAS [4] and show that it fails to achieve *pre-verify soundness* and satisfies neither standard nor our redefined *weak unlinkability*. In a standard signature, the response $\mathbf{z}$ is uniformly distributed subject to $\|\mathbf{z}\|_\infty \leq \gamma - \kappa$ (denoted $\mathcal{D}_0$), whereas an adapted signature uses $\mathbf{z} = \hat{\mathbf{z}} + \mathbf{y}$, where $\mathbf{y}$ is uniformly drawn subject to $\|\mathbf{y}\|_\infty \leq 1$ (denoted $\mathcal{D}_1$). Crucially, the probability of a coefficient hitting the upper boundary fundamentally differs between these two distributions: $p_0 = 1/(2(\gamma - \kappa) + 1)$ in $\mathcal{D}_0$, while $p_1 = 1/(3(2(\gamma - \kappa) - 1))$ in $\mathcal{D}_1$. We present an explicit distinguishing algorithm exploiting this boundary-density variance. By collecting a polynomial number of signatures, a probabilistic polynomial-time (PPT) adversary can distinguish adapted signatures from standard ones with non-negligible advantage. Both *unlinkability* and our *weak unlinkability* require these signatures to be computationally indistinguishable. Ultimately, the failure of LAS to meet these requirements allows a passive observer to identify signatures generated via the adaptor mechanism, thereby compromising privacy in applications such as atomic swaps. Moreover, LAS lacks *pre-verify soundness*. In lattice settings, it is computationally hard to determine whether a public statement Y possesses a valid

small-norm witness $\mathbf{y}$. Since LAS lacks mechanisms to verify statement well-formedness, a malicious counterparty can inject invalid statements without short preimages, rendering final witness extraction impossible and breaking protocol fairness.

- **The LAS⁺ Construction.** To remedy these gaps, we present LAS⁺. Before detailing our approach, we clarify our target security model. Achieving standard unlinkability is inherently challenging in lattices due to the strict short-norm witness requirements. To properly address this, we formalize *weak unlinkability*, where the challenge relation is generated honestly by the challenger rather than adaptively chosen by the adversary. Importantly, this weak notion fully guarantees on-chain privacy against external observers, since such observers monitor the ledger passively and cannot maliciously dictate cryptographic relations.

LAS⁺ realizes *weak unlinkability* and *pre-verify soundness* through two core enhancements. First, we introduce a distribution-adjusting rejection-sampling mechanism. We treat the original response distribution as a proposal distribution g and the adapted-signature distribution as the target f . Given a candidate response $\mathbf{z}$, the signer accepts it with probability $f(\mathbf{z})/(Mg(\mathbf{z}))$ for a normalizing constant $M \geq \sup_{\mathbf{z}} f(\mathbf{z})/g(\mathbf{z})$. The resulting output exactly matches the adapted-signature distribution, restoring weak unlinkability. By tuning the parameters, candidate signatures are accepted on the first attempt with high probability. Second, to achieve *pre-verify soundness*, we expand the public statement Y into a composite statement $\mathbf{Y} = (Y, \pi)$, where π is an attached NIZK proof. This enables the verifier to reject any malformed statement Y , successfully blocking malicious injections at the entry point.

- **Security Proofs under the Stronger Model.** We formally prove that LAS⁺ satisfies the strengthened *extractability* and *unique extractability* defined in [6]. The stronger security model introduces new challenges, as *extractability* now subsumes both unforgeability and witness extractability, allowing the adversary to make multiple adaptive PreSign and NewY (relation sampling) queries. We overcome this by reducing a successful forgery to solving the Module-SIS (M-SIS) assumption.

We perform a structured case analysis on how the forged signature $(c, \mathbf{z})$ relates to the oracle transcripts: (i) If the forgery reuses a challenge c from a PreSign query but is not tied to any NewY query, it yields a valid witness but directly contradicts the game’s winning condition. (ii) If the forgery is linked to both queries, the simulator embeds the M-SIS challenge matrix $\mathbf{B} = [\mathbf{I}_n \parallel \mathbf{A}' \parallel \mathbf{a}]$ (where $\mathbf{I}_n$ is the identity matrix, and $\mathbf{A}', \mathbf{a}$ are uniformly random) by setting the public parameters to $\mathbf{A} = [\mathbf{I}_n \parallel \mathbf{A}']$ and returning statements $Y_i = \mathbf{A}\mathbf{r}_i - \mathbf{a}$ via the NewY oracle. A matching forgery then yields an extracted short kernel vector $\mathbf{s}' = (\mathbf{z} - \hat{\mathbf{z}} - \mathbf{r}_i, 1)^\top$ that satisfies $\mathbf{B}\mathbf{s}' = \mathbf{0} \bmod q$, successfully solving the M-SIS instance. (iii) If the forgery is not linked to any pre-signature, a standard rewinding argument produces two valid signatures on the same statement, whose difference yields a short kernel vector.

Finally, *unique extractability* follows naturally: if a single pre-signature leads to two distinct valid signatures $(c_0, \mathbf{z}_0)$ and $(c_1, \mathbf{z}_1)$, their algebraic difference gives $\mathbf{A}(\mathbf{z}_0 - \mathbf{z}_1) = \mathbf{0}$, directly yielding a non-trivial short solution to M-SIS.

2 Preliminaries

We use λ to denote the security parameter. We work over the cyclotomic ring $R_q = \mathbb{Z}_q[X]/(X^d + 1)$ of power-of-two degree d (for odd q), and write $R = \mathbb{Z}[X]/(X^d + 1)$. For $c \in \mathbb{Z}^+$, let $\mathcal{S}_c$ denote the set of polynomials in R_q whose coefficients have absolute value at most c . Vectors and matrices over these rings are written in bold (e.g., $\mathbf{a}, \mathbf{A}$), and $\mathbf{I}_n$ denotes the n -dimensional identity matrix. For a polynomial $f = \sum_{i=0}^{d-1} f_i X^i \in R$, we use the standard norms $\|f\|_\infty = \max_i |f_i|$, and $\|f\|_1 = \sum_i |f_i|$. We write $x \xleftarrow{\$} X$ for sampling x uniformly from a finite set X , and $y \leftarrow A(x; r)$ for the output of a probabilistic polynomial-time (PPT) algorithm A on input x and randomness r . For a deterministic polynomial-time (DPT) algorithm A , we use $y := A(x)$. A function $\nu : \mathbb{N} \rightarrow \mathbb{R}$ is negligible if for every $k \in \mathbb{N}$ there exists n_0 such that $|\nu(n)| \leq n^{-k}$ for all $n \geq n_0$. We also use the command **assert** to enforce conditions during games: if a stated condition is false, the game aborts.

Definition 1 (Hard Relation). *A relation R with language $L_R := \{Y \mid \exists y : (Y, y) \in R\}$ is hard if there exists a PPT generator $R.\text{Gen}(1^\lambda)$ that outputs $(Y, y) \in R$, and for any PPT adversary $\mathcal{A}$, the probability $\Pr[\mathcal{A}(1^\lambda, Y) = y'] \leq \nu(\lambda)$, where $\nu(\lambda)$ is negligible.*

Definition 2 (Digital Signature). *A signature scheme $\Sigma = (\text{KGen}, \text{Sign}, \text{Verify})$ consists of three algorithms:*

- $(\text{sk}, \text{pk}) \leftarrow \text{KGen}(1^\lambda)$: on input a security parameter λ , the PPT algorithm KGen outputs public key pk and secret key sk .
- $\sigma \leftarrow \text{Sign}(\text{sk}, m)$: on input a secret key sk , message $m \in \{0, 1\}^*$, the PPT algorithm Sign outputs a signature σ .
- $b := \text{Verify}(\text{pk}, m, \sigma)$: on input a public key pk , message $m \in \{0, 1\}^*$ and signature σ , the DPT algorithm Verify outputs a bit b .

The security of our scheme relies on the standard Module-SIS (M-SIS) and Module-LWE (M-LWE) hardness assumptions over the ring R_q . Specifically, we rely on the Hermite Normal Form (HNF) of the M-SIS problem, which tasks an adversary with finding a short non-zero vector $\mathbf{v}$ such that $[\mathbf{I}_n \parallel \mathbf{A}']\mathbf{v} = \mathbf{0} \bmod q$. For M-LWE, we adopt the widely used small-secret variant (analogous to Dilithium [3]), where distinguishing the noisy product $\mathbf{As} + \mathbf{e}$ from a uniformly random vector is computationally intractable. Due to space constraints, we defer the formal definitions and parameter details of these standard assumptions to Appendix A.1.

2.1 Adaptor Signature

Definition 3 (Adaptor signature [4]). An adaptor signature scheme $AS_{\Sigma, R, R'}$, defined with respect to a base signature scheme $\Sigma = (\text{KGen}, \text{Sign}, \text{Verify})$ and two hard relations $R \subseteq R'$, consists of a tuple of four algorithms $AS_{\Sigma, R, R'} = (\text{PreSign}, \text{PreVerify}, \text{Adapt}, \text{Extract})$ defined as:

- $\tilde{\sigma} \leftarrow \text{PreSign}(\text{sk}, m, Y)$: on input a secret key sk , message $m \in \{0, 1\}^*$ and statement $Y \in L_R$, the PPT algorithm PreSign outputs a pre-signature $\tilde{\sigma}$.
- $b := \text{PreVerify}(\text{pk}, m, Y, \tilde{\sigma})$: on input a public key pk , message $m \in \{0, 1\}^*$, statement $Y \in L_R$ and pre-signature $\tilde{\sigma}$, the DPT algorithm PreVerify outputs a bit b .
- $\sigma \leftarrow \text{Adapt}(\text{pk}, \tilde{\sigma}, y)$: on input a pre-signature $\tilde{\sigma}$ and witness y for the statement $Y \in L_R$, the PPT algorithm Adapt outputs an adapted signature σ .
- $y := \text{Extract}(\text{pk}, \tilde{\sigma}, \sigma, Y)$: on input a pre-signature $\tilde{\sigma}$, signature σ and statement $Y \in L_R$, the DPT algorithm Extract outputs a witness y such that $(Y, y) \in R'$, or $\perp$.

The distinction between R and R' reflects a well-known “knowledge gap” in lattice-based zero-knowledge proofs. Because reductions often rely on the forking lemma and involve rewinding combined with rejection sampling, the extracted witness may not follow the same distribution as an honestly generated one. Thus, honest executions naturally produce witnesses in the strict relation R , while extraction in security proofs may recover witnesses only satisfying the relaxed relation R' . Lattice-based adaptor signatures inherit this gap, motivating the formalization of security with respect to two relations $R \subseteq R'$. We often omit the subscript from $AS_{\Sigma, R, R'}$ writing AS to improve readability if the signature scheme Σ and the hard relations R, R' are clear from the context.

Definition 4 ((Weak) Pre-signature correctness [4]).

An adaptor signature scheme AS satisfies weak pre-signature correctness if for every $\lambda \in \mathbb{N}$, $m \in \{0, 1\}^*$, and $(Y, y) \in R$, generating $(\text{sk}, \text{pk}) \leftarrow \text{KGen}(1^\lambda)$, $\tilde{\sigma} \leftarrow \text{PreSign}(\text{sk}, m, Y)$, $\sigma \leftarrow \text{Adapt}(\text{pk}, \tilde{\sigma}, y)$, and $y' \leftarrow \text{Extract}(\text{pk}, \tilde{\sigma}, \sigma, Y)$ yields:

$$\Pr[\text{PreVerify}(\text{pk}, m, Y, \tilde{\sigma}) = 1 \wedge \text{Verify}(\text{pk}, m, \sigma) = 1 \wedge (Y, y') \in R'] = 1.$$

Definition 5 ((Weak) Pre-signature adaptability [4]). An adaptor signature scheme AS satisfies weak pre-signature adaptability if for any $\lambda \in \mathbb{N}$, any message $m \in \{0, 1\}^*$, any statement/witness pair $(Y, y) \in R$, any key pair $(\text{sk}, \text{pk}) \leftarrow \text{KGen}(1^\lambda)$ and any pre-signature $\tilde{\sigma} \in \{0, 1\}^*$ with $\text{PreVerify}(\text{pk}, m, Y, \tilde{\sigma}) = 1$, we have $\Pr[\text{Verify}(\text{pk}, m, \text{Adapt}(\text{pk}, \tilde{\sigma}, y)) = 1] = 1$.

We call our notion of pre-signature adaptability *weak* because it guarantees adaptability only for honest witnesses in the original relation R , rather than the relaxed relation R' . As noted earlier, this distinction is inherent to the lattice-based knowledge gap. Thus, unlike classical adaptor signatures where $R = R'$ and full adaptability holds, the lattice-based setting admits a separation between the two relations.

2.2 Stronger Security Model for Adaptor Signatures

Classical security models for adaptor signatures focus mainly on aEUF-CMA security together with witness extractability. The recent work [6] refines this view by adopting a stronger model that strengthens the adversary (allowing multiple adaptive pre-signature queries) and adds further requirements such as *extractability*, *unique extractability*, *unlinkability*, and *pre-verify soundness*.

Definition 6 (Extractability). *An adaptor signature scheme AS is extractable, if for every PPT adversary $\mathcal{A}$, there exists a negligible function ν such that for every $\lambda \in \mathbb{N}$, $\Pr[\text{Ext}_{\mathcal{A},\text{AS}}(\lambda) = 1] \leq \nu(\lambda)$, where the experiment $\text{Ext}_{\mathcal{A},\text{AS}}$ is described below, and the probability is taken over the random choices of all probabilistic algorithms.*

$\text{Ext}_{\mathcal{A},\text{AS}}(\lambda)$	$\text{Sign}(\text{sk}, m)$
1: $(\text{sk}, \text{pk}) \leftarrow \text{KGen}(1^\lambda); \quad b \leftarrow 1; \quad \mathcal{S}, \mathcal{T} \leftarrow \emptyset$	1: $\sigma \leftarrow \text{Sign}(\text{sk}, m)$
2: $(m^*, \sigma^*) \leftarrow \mathcal{A}^{\text{NewY}(\lambda), \text{Sign}(\text{sk}, \cdot), \text{PreSign}(\text{sk}, \cdot, \cdot)}(\text{pk})$	2: $\mathcal{S} \leftarrow \mathcal{S} \cup \{m\}$
3: assert $\text{Verify}(\text{pk}, m^*, \sigma^*)$	3: return σ
4: assert $m^* \notin \mathcal{S}$	$\text{PreSign}(\text{sk}, m, Y)$
5: for $(Y, \tilde{\sigma}) \in \mathcal{T}[m^*]$ do	1: $\tilde{\sigma} \leftarrow \text{PreSign}(\text{sk}, m, Y)$
6: if $(Y, \text{Extract}(\text{pk}, \tilde{\sigma}, \sigma^*, Y)) \in \mathbf{R}'$ then $b \leftarrow 0$	2: $\mathcal{T}[m] \leftarrow \mathcal{T}[m] \cup \{(Y, \tilde{\sigma})\}$
7: return b	3: return $\tilde{\sigma}$
	$\text{NewY}(\lambda)$
	1: $(Y, y) \leftarrow \text{R.Gen}(1^\lambda)$
	2: return Y

Extractability thus ensures both unforgeability and meaningful binding to the relation, requiring that every valid signature correspond to a witness that can be efficiently extracted from its pre-signature and preventing any witness-free adaptation. Unlike prior definitions, where the adversary could see only a single pre-signature on the challenge message, the extractability experiment allows the adversary to obtain multiple pre-signatures on the same challenge as well as on multiple honestly generated statements. In classical adaptor-signature schemes one typically has $\mathbf{R}' = \mathbf{R}$; in our lattice setting we instead work with the relaxed relation $\mathbf{R}'$ introduced above, thereby extending the classical notion of extractability.

Definition 7 (Unique Extractability). *An adaptor signature scheme AS is unique extractable, if for every PPT adversary $\mathcal{A}$ there exists a negligible function ν such that for every $\lambda \in \mathbb{N}$, $\Pr[\text{UniqueExtractability}_{\mathcal{A},\text{AS}}(\lambda) = 1] \leq \nu(\lambda)$, where experiment $\text{UniqueExtractability}_{\mathcal{A},\text{AS}}$ is described below, and the probability is taken over the random choices of all probabilistic algorithms.*

UniqueExtractability _{A,AS} (λ)	Sign(sk, m)
1 : (sk, pk) ← KGen(1 ^λ)	1 : σ ← Sign(sk, m)
2 : (m, Y, σ̃, σ, σ') ← A ^{Sign(sk, ·), PreSign(sk, ·, ·)} (pk)	2 : return σ
3 : assert (σ ≠ σ') ∧ Verify(pk, m, σ) ∧ Verify(pk, m, σ')	PreSign(sk, m, Y)
4 : assert PreVerify(pk, m, Y, σ̃)	1 : σ̃ ← PreSign(sk, m, Y)
5 : y ← Extract(pk, σ̃, σ, Y); y' ← Extract(pk, σ̃, σ', Y)	2 : return σ̃
6 : return ((y ≠ y') ∧ (Y, y) ∈ R' ∧ (Y, y') ∈ R')	

Unique extractability enforces a one-to-one mapping from a pre-signature to a valid full signature, yielding a unique witness. In our lattice setting, this uniqueness is interpreted with respect to the relaxed relation R' , rather than the strict R assumed in classical definitions.

Definition 8 (Weak Unlinkability). *An adaptor signature scheme AS is weak unlinkable, if for every PPT adversary $\mathcal{A}$ there exists a negligible function ν such that for every $\lambda \in \mathbb{N}$, $\left| \Pr[\text{WeakUnlink}_{\mathcal{A},\text{AS}}(\lambda, 0) = 1] - \Pr[\text{WeakUnlink}_{\mathcal{A},\text{AS}}(\lambda, 1) = 1] \right| \leq \nu(\lambda)$, where experiment $\text{WeakUnlink}_{\mathcal{A},\text{AS}}$ is described below, and the probability is taken over the random choices of all probabilistic algorithms.*

WeakUnlink _{A,AS} (λ, b)	Chall(b, sk, m)
1 : (sk, pk) ← KGen(1 ^λ)	1 : (Y, y) ← R.Gen(1 ^λ)
2 : b' ← A ^{Chall(b, sk, ·), Sign(sk, ·), PreSign(sk, ·, ·)} (pk)	2 : σ̃ ← PreSign(sk, m, Y)
3 : return b'	3 : σ ₀ ← Sign(sk, m)
PreSign(sk, m, Y)	4 : σ ₁ := Adapt(pk, σ̃, y)
1 : σ̃ ← PreSign(sk, m, Y)	5 : return σ _b
2 : return σ̃	Sign(sk, m)
	1 : σ ← Sign(sk, m)
	2 : return σ

Weak unlinkability ensures that standard and adapted signatures remain computationally indistinguishable, preserving on-chain privacy (e.g., in atomic swaps) against external observers. Unlike standard unlinkability, where the adversary adaptively chooses the hard relation (Y, y) , weak unlinkability requires the challenger to honestly generate it. We adopt this relaxed notion for our LAS^+ construction for two reasons. First, it still aligns with practical scenarios such as atomic swaps: external observers cannot control relation generation, and transaction participants inherently know the linkage. Thus, granting the adversary the power to choose (Y, y) is unnecessarily strong for such use cases. Second, achieving standard unlinkability is inherently challenging in the lattice setting because the adversary could manipulate the distribution of the witness y while satisfying the short norm requirement, and thereby control the distribution of

the final adapted signature. Consequently, efficiently masking the distribution manipulation caused by adversary-chosen relations in the lattice setting remains open for future work.

Definition 9 (Computational Pre-Verify Soundness). *An adaptor signature scheme AS satisfies computational pre-verify soundness if for every PPT adversary $\mathcal{A}$ there exists a negligible function ν such that, for every $\lambda \in \mathbb{N}$ and polynomially-bounded $Y \notin L_R$,*

$$\Pr \left[(\text{sk}, \text{pk}) \leftarrow \text{KGen}(1^\lambda); (m, \tilde{\sigma}) \leftarrow \mathcal{A}(\text{sk}, Y) : \text{PreVerify}(\text{pk}, m, Y, \tilde{\sigma}) = 1 \right] \leq \nu(\lambda).$$

Pre-verify soundness requires that the pre-verification algorithm accept only those pre-signatures associated with statements Y that fall in the relation. This property ensures that no malformed or invalid statement can pass pre-verification, thereby complementing pre-signature adaptability.

3 Revisiting the LAS Scheme

In this section, we revisit the LAS scheme [4] and analyze its security under the stronger security model as defined in Section 2.2. Our analysis reveals that the original LAS scheme fails to achieve unlinkability (even the weak unlinkability). Moreover, we observe that the LAS scheme does not inherently support pre-verify soundness. This motivates us to refine the LAS design so as to achieve stronger security in Section 4.

3.1 LAS scheme

The LAS scheme [4] consists of two integral components: (i) a standard lattice-based signature scheme and (ii) an adaptor signature extension built on top of it. The underlying lattice-based signature follows the structure of the simplified Dilithium scheme [3]. The main identifiers and parameter choices are listed in Table 1, and we outline the full set of algorithms below.

The Standard Signature of LAS.

- $\text{KGen}(\lambda)$: Sample $\mathbf{r} \xleftarrow{\$} \mathcal{S}_1^{n+\ell}$. Return the key pair $(\text{sk}, \text{pk}) = (\mathbf{r}, \mathbf{A}\mathbf{r})$.
- $\text{Sign}(\text{sk}, m)$: Sample $\mathbf{u} \xleftarrow{\$} \mathcal{S}_\gamma^{n+\ell}$. Compute $c = H(\text{pk}, \mathbf{A}\mathbf{u}, m)$ and $\mathbf{z} = \mathbf{u} + c \cdot \text{sk}$. Return the signature $\sigma = (c, \mathbf{z})$ if $\|\mathbf{z}\|_\infty \leq \gamma - \kappa$; else restart.
- $\text{Verify}(\text{pk}, m, \sigma)$: Parse σ as $(c, \mathbf{z})$. Return 1 if $\|\mathbf{z}\|_\infty \leq \gamma - \kappa$ and $c = H(\text{pk}, \mathbf{A}\mathbf{z} - c \cdot \text{pk}, m)$; else 0.

The Adaptor Signature of LAS.

- $\text{PreSign}(\text{sk}, m, Y)$: Sample $\mathbf{u} \xleftarrow{\$} \mathcal{S}_\gamma^{n+\ell}$. Compute $c = H(\text{pk}, \mathbf{A}\mathbf{u} + Y, m)$ and $\hat{\mathbf{z}} = \mathbf{u} + c \cdot \text{sk}$. Return the pre-signature $\tilde{\sigma} = (c, \hat{\mathbf{z}})$ if $\|\hat{\mathbf{z}}\|_\infty \leq \gamma - \kappa - 1$; else restart.

- $\text{PreVerify}(\text{pk}, m, Y, \tilde{\sigma})$: Parse $\tilde{\sigma}$ as $(c, \hat{\mathbf{z}})$. Return 1 if $\|\hat{\mathbf{z}}\|_\infty \leq \gamma - \kappa - 1$ and $c = H(\text{pk}, \mathbf{A}\hat{\mathbf{z}} - c \cdot \text{pk} + Y, m)$; else 0.
- $\text{Adapt}(\text{pk}, \tilde{\sigma}, y)$: Parse $\tilde{\sigma} = (c, \hat{\mathbf{z}})$. Return the adapted full signature $\sigma = (c, \hat{\mathbf{z}} + \mathbf{y})$ if $\text{PreVerify}(\text{pk}, m, Y, \tilde{\sigma}) = 1$; else $\perp$.
- $\text{Extract}(\text{pk}, \tilde{\sigma}, \sigma, Y)$: Parse $\sigma = (c, \mathbf{z})$ and $\tilde{\sigma} = (c, \hat{\mathbf{z}})$. Compute $\mathbf{s} = \mathbf{z} - \hat{\mathbf{z}}$. Return the extracted witness $\mathbf{s}$ if $\mathbf{A}\mathbf{s} = Y$; else $\perp$.

Table 1. Notation and parameter choices for LAS and LAS⁺.

Notation	Explanation	Value
d	ring dimension (a power of two)	256
$\mathcal{R}_q$	cyclotomic ring of degree d , $\mathcal{R}_q = \mathbb{Z}_q[X]/(X^d + 1)$	$\log q \approx 24$
$\mathcal{S}_c$	polynomials $f \in \mathcal{R}_q$ with $\ f\ _\infty \leq c$, for $c \in \mathbb{Z}^+$	—
n	rank of the underlying M-SIS instance	4
ℓ	rank of the underlying M-LWE instance	4
$\mathbf{A}$	public matrix in systematic form $[\mathbf{I}_n \parallel \mathbf{A}']$, with $\mathbf{A}' \xleftarrow{\$} \mathcal{R}_q^{n \times \ell}$	$\mathcal{R}_q^{n \times (n+\ell)}$
$\mathcal{C}$	challenge space (image of $\mathbf{H}$): $\{c \in \mathcal{R} : \ c\ _1 = \kappa, \ c\ _\infty = 1\}$	$\kappa = 60$
γ	upper bound on coefficients of the masking vector	$\kappa d(n + \ell)$

The LAS scheme operates over the standard base relation $\mathfrak{R}_{\mathbf{A}} = \{(Y, \mathbf{y}) \mid \mathbf{A}\mathbf{y} = Y \bmod q, \|\mathbf{y}\|_\infty \leq 1\}$ and its relaxed version $\mathfrak{R}'_{\mathbf{A}}$ where the bound is relaxed to $\|\mathbf{y}\|_\infty \leq 2(\gamma - \kappa)$.

3.2 The Unlinkability Gap

We next explain the intuition behind our unlinkability analysis. Recall from Lemma 4.3 of [8] that in the Fiat–Shamir with Aborts procedure, the response vector $\mathbf{z}$ of a standard signature is uniformly distributed over the hypercube $\mathcal{S}_{\gamma-\kappa}^{n+\ell}$. Thus, every vector in $\mathcal{S}_{\gamma-\kappa}^{n+\ell}$ is output with the same probability, and the mass on boundary coefficients $\pm(\gamma - \kappa)$ is relatively large.

In contrast, an adapted signature in LAS is obtained by adding a short witness to a pre-signature. Concretely, the pre-signature response $\hat{\mathbf{z}}$ is sampled (after rejection) from $\mathcal{S}_{\gamma-\kappa-1}^{n+\ell}$ and combined with a witness $\mathbf{y} \xleftarrow{\$} \mathcal{S}_1^{n+\ell}$ to form $\mathbf{z} = \hat{\mathbf{z}} + \mathbf{y}$. This yields a convolution of two short distributions, producing a vector where the two outermost coefficient values receive strictly less mass. Intuitively, standard signatures attain the boundary value with higher probability.

We now view this discrepancy through the unlinkability experiment. The challenger samples a hidden bit $b \in \{0, 1\}$ and returns k signatures $\sigma_i = (c_i, \mathbf{z}_i)$ to the adversary $\mathcal{A}$, where all $\mathbf{z}_i$ are generated according to one of the following distributions: $\mathcal{D}_0 : \mathbf{z} \xleftarrow{\$} \mathcal{S}_{\gamma-\kappa}^{n+\ell}$ or $\mathcal{D}_1 : \mathbf{z} = \hat{\mathbf{z}} + \mathbf{y}$ where $\hat{\mathbf{z}} \xleftarrow{\$} \mathcal{S}_{\gamma-\kappa-1}^{n+\ell}, \mathbf{y} \xleftarrow{\$} \mathcal{S}_1^{n+\ell}$. Unlinkability requires $\mathcal{D}_0$ and $\mathcal{D}_1$ to be computationally indistinguishable.

To show that this discrepancy violates unlinkability for LAS, we focus on the probability that a single coefficient attains the upper boundary $\gamma - \kappa$. Let p_0 and p_1 be this probability under $\mathcal{D}_0$ and $\mathcal{D}_1$, respectively. A direct calculation yields $p_0 = 1/(2(\gamma - \kappa) + 1)$ and $p_1 = 1/(3(2(\gamma - \kappa) - 1))$, so that the gap $\Delta := p_0 - p_1 > 0$. An adversary can therefore aggregate all coefficients from the k signatures, count how many lie on the boundary, and compare this count to a threshold T chosen according to p_0 and p_1 . If the count exceeds T , it outputs $b' = 0$; otherwise, $b' = 1$. By a standard Hoeffding bound, this test distinguishes $\mathcal{D}_0$ from $\mathcal{D}_1$ with non-negligible advantage, breaking the unlinkability of LAS.

Success Probability and Advantage. We analyze the distinguishing advantage of the adversary $\mathcal{A}$ by evaluating its decision error probabilities. Specifically, let α denote the false positive rate (i.e., deciding $\mathcal{D}_1$ when the true distribution is $\mathcal{D}_0$), and β denote the false negative rate (i.e., deciding $\mathcal{D}_0$ when it is actually $\mathcal{D}_1$). The adversary's overall advantage is then given by $\text{Adv}_{\mathcal{A}} = 1 - (\alpha + \beta)$.

Adversary Strategy $\mathcal{A}^{\text{Chall}(b, \text{sk}, \cdot), \text{Sign}(\text{sk}, \cdot), \text{PreSign}(\text{sk}, \cdot, \cdot)}$	
1 :	Receive k signatures : $\sigma_1, \dots, \sigma_k$
2 :	Extract all response vectors z_i from each σ_i
3 :	Collect all coefficients into a multiset;
4 :	Set $N \leftarrow k \cdot (n + \ell) \cdot d$
5 :	for $i = 1$ to N do
6 :	Define $X_i \leftarrow 1$ if coefficient $i = \gamma - \kappa$, else 0
7 :	endfor
8 :	$K \leftarrow \sum_{i=1}^N X_i$; Set a threshold $T = N \cdot \frac{p_0 + p_1}{2}$
9 :	if $K \geq T$ then
10 :	return $\hat{b} \leftarrow 0$
11 :	else
12 :	return $\hat{b} \leftarrow 1$
13 :	endif

Let $X_1, \dots, X_N$ be i.i.d. Bernoulli variables where each X_i indicates whether a sampled coefficient hits the upper boundary, with its mean being $\mathbb{E}[X_i] = p_b$ for $b \in \{0, 1\}$. We define the total boundary count as $K = \sum_{i=1}^N X_i$, the decision threshold as the midpoint $T := N(p_0 + p_1)/2$, and the probability gap as $\Delta := p_0 - p_1 > 0$. Under $\mathcal{D}_0$, the expected count is Np_0 , so a decision error (α) occurs if $K < T$. Conversely, under $\mathcal{D}_1$, the expected count is Np_1 , and an error (β) occurs if $K \geq T$. In both scenarios, an error implies that K must deviate from its mean by at least $N\Delta/2$. Applying Hoeffding's inequality with a deviation margin of $N\Delta/2$, we can bound the directional error probabilities strictly by $\alpha = \Pr_{\mathcal{D}_0}[K < T] \leq \exp(-N\Delta^2/2)$ and $\beta = \Pr_{\mathcal{D}_1}[K \geq T] \leq \exp(-N\Delta^2/2)$. Consequently, the sum of errors is bounded by $\alpha + \beta \leq 2 \exp(-N\Delta^2/2)$, establishing the lower

bound for the advantage as:

$$\text{Adv}_{\mathcal{A}} \geq 1 - 2 \exp\left(-\frac{N\Delta^2}{2}\right).$$

Since $\Delta > 0$ and $N = \text{poly}(\lambda)$, the adversary gains a non-negligible advantage in distinguishing adapted signatures from standard ones. Therefore, the scheme does not satisfy unlinkability. Furthermore, it also fails *weak unlinkability*, which similarly requires standard and adapted signatures to be indistinguishable. For a concrete evaluation using the parameters in Table 1 ($n = \ell = 4$, $d = 256$, $\kappa = 60$), the probabilities of hitting the boundary are $p_0 \approx 4.07 \times 10^{-6}$ and $p_1 \approx 1.36 \times 10^{-6}$, yielding a gap of $\Delta \approx 2.71 \times 10^{-6}$. Since each signature leaks $(n + \ell)d = 2048$ coefficients, an observer accumulating $k \approx 4 \times 10^8$ on-chain signatures pools $N \approx 8.1 \times 10^{11}$ samples and thus attains $\text{Adv}_{\mathcal{A}} \geq 0.9$.

3.3 Absence of Pre-verify Soundness Formalization

The *pre-verify soundness* property requires that any pre-signature generated with an invalid statement Y (i.e., one lacking a valid witness) must be rejected during pre-verification. In discrete-log settings, verifying the well-formedness of Y is typically straightforward, such as simply checking subgroup membership. However, in lattice-based cryptography, determining whether a public statement Y genuinely admits a valid small-norm preimage y given only Y is computationally hard. In fact, distinguishing valid statements with short preimages from uniformly random elements is intrinsically tied to the hardness of the underlying lattice problems. Consequently, without auxiliary information, a verifier cannot efficiently detect and reject malformed statements.

The original LAS security model did not formally capture this requirement, as its core syntax lacks a built-in mechanism to certify the small-norm witness relation. Indeed, Esgin et al. [4] recognized this challenge in practice: when deploying LAS to their atomic swap application, they mandated that participants attach a zero-knowledge proof to guarantee that the extracted witness y is valid and strictly satisfies the required norm bounds. We draw direct inspiration from this application-level design. We abstract this idea and formally integrate the zero-knowledge proof requirement into the core syntax and security definitions of the adaptor signature. This integration ensures that our scheme achieves pre-verify soundness.

4 LAS⁺: Lattice-based Adaptor Signature with Stronger Security

In this section, we first introduce the rejection sampling theorem similar to [10] but customized for our parameter sets. Building upon this theorem, we present our enhanced construction LAS⁺, which addresses the unlinkability weakness of the original LAS scheme. Furthermore, we explicitly integrate a non-interactive

LAS ⁺ Scheme	
Standard signature	
Sign(sk, m)	
1 : $\mathbf{u} \xleftarrow{\$} \mathcal{S}_\gamma^{n+\ell}$ 2 : $\mathbf{w} = \mathbf{A}\mathbf{u}$ 3 : $c = H(\text{pk}, \mathbf{w}, m)$ 4 : $\mathbf{z} = \mathbf{u} + c \cdot \text{sk}$ 5 : if $\ \mathbf{z}\ _\infty > \gamma - \kappa$ then restart 6 : $U \xleftarrow{\\$} [0, 1]$ restart if $U > \frac{f(\mathbf{z})}{M \cdot g(\mathbf{z})}$ 7 : return $\sigma = (c, \mathbf{z})$	
Adaptor signature	
PreSign(sk, m, Y)	PreVerify(pk, m, Y, $\tilde{\sigma}$)
1 : $\text{Parse}(Y, \pi) := \mathbf{Y}$ if NIZK.V(Y, π) = 0 then return $\perp$ 2 : $\mathbf{u} \xleftarrow{\$} \mathcal{S}_\gamma^{n+\ell}$ 3 : $\mathbf{w} = \mathbf{A}\mathbf{u}$ 4 : $c = H(\text{pk}, \mathbf{w} + Y, m)$ 5 : $\hat{\mathbf{z}} = \mathbf{u} + c \cdot \text{sk}$ 6 : if $\ \hat{\mathbf{z}}\ _\infty > \gamma - \kappa - 1$ 7 : then restart 8 : return $\tilde{\sigma} = (c, \hat{\mathbf{z}})$	1 : $\text{Parse}(Y, \pi) := \mathbf{Y}$ if NIZK.V(Y, π) = 0 then return 0 2 : $\text{Parse}(c, \hat{\mathbf{z}}) := \tilde{\sigma}$ 3 : if $\ \hat{\mathbf{z}}\ _\infty > \gamma - \kappa - 1$ 4 : then return 0 5 : $\mathbf{w}' = \mathbf{A}\hat{\mathbf{z}} - c \cdot \text{pk}$ 6 : if $c \neq H(\text{pk}, \mathbf{w}' + Y, m)$ 7 : then return 0 8 : return 1

Fig. 1. The LAS⁺ signature scheme (only modified algorithms are shown).
The highlighted parts indicate the modifications introduced in LAS⁺.

zero-knowledge (NIZK) proof system into the adaptor signature protocol to guarantee pre-verify soundness under the lattice setting. Finally, we formally prove that LAS⁺ satisfies weak unlinkability, pre-verify soundness, extractability, and unique extractability as defined in the stronger security model in Section 2.2.

4.1 LAS⁺ Scheme

The structure of LAS⁺ largely follows that of the original LAS scheme, combining a lattice-based signature with an adaptor-signature extension. Specifically, the underlying parameters are chosen as shown in Table 1, and the base hard relation is defined as in Section 3.1. However, as discussed earlier, this standard base

relation does not align with pre-verify soundness since, given a public statement Y , it is computationally hard to determine whether Y belongs to the relation language by verifying the existence of a corresponding short-norm witness y .

To resolve this issue, LAS^+ introduces an NIZK-augmented relation. The NIZK proof can be instantiated using the scheme proposed in [5], whose soundness guarantees that $\|y\|_\infty \leq 1$ holds for $Y = \mathbf{A}y$. By elevating the public statement to a composite $\mathbf{Y} = (Y, \pi)$, where π explicitly certifies the valid short preimage, the augmented base relation $\hat{\mathfrak{R}}_{\mathbf{A}}$ and extended relation $\hat{\mathfrak{R}}'_{\mathbf{A}}$ are defined as:

$$\begin{aligned}\hat{\mathfrak{R}}_{\mathbf{A}} &= \{(\mathbf{Y}, y) \mid \mathbf{Y} = (Y, \pi) \wedge \mathbf{A}y = Y \bmod q \wedge \|y\|_\infty \leq 1 \wedge \text{NIZK.V}(Y, \pi) = 1\}, \\ \hat{\mathfrak{R}}'_{\mathbf{A}} &= \{(\mathbf{Y}, y) \mid \mathbf{Y} = (Y, \pi) \wedge \mathbf{A}y = Y \bmod q \wedge \|y\|_\infty \leq 2(\gamma - \kappa) \wedge \text{NIZK.V}(Y, \pi) = 1\},\end{aligned}$$

so that $\hat{\mathfrak{R}}_{\mathbf{A}} \subseteq \hat{\mathfrak{R}}'_{\mathbf{A}}$. Building upon this foundation, our construction introduces two essential modifications to address the limitations of LAS under the stronger security model.

First, we incorporate a rejection-sampling step within the **Sign** algorithm. This ensures that a standard signature remains indistinguishable from the adapted signature, thus satisfying *weak unlinkability*.

Second, LAS^+ upgrades the underlying protocol to operate over the NIZK-augmented relations $\hat{\mathfrak{R}}_{\mathbf{A}}$ and $\hat{\mathfrak{R}}'_{\mathbf{A}}$, which expands the public statement into a composite $\mathbf{Y} = (Y, \pi)$. Leveraging this structure, we explicitly integrate the corresponding non-interactive zero-knowledge proof check $\text{NIZK.V}(Y, \pi) = 1$ into both the **PreSign** and **PreVerify** algorithms. By mandating this verification to certify the well-formedness of the base statement Y , this enhancement successfully guarantees *pre-verify soundness*. Next, we explain our techniques in detail. **Rejection-sampling refinement.** In the **Sign** algorithm, we add a tuned rejection sampling step to reshape the response distribution $\mathbf{z} = \mathbf{u} + c \cdot \text{sk}$ so that it matches the distribution of adapted signatures. This ensures that standard and adapted signatures become identically distributed, restoring the unlinkability property that the original LAS fails to achieve under the stronger security model. All other procedures, including key generation, verification and the algorithms for adaptor signatures, remain unchanged. The modified structure of LAS^+ is shown in Figure 1, with the rejection-sampling step highlighted. This refinement guarantees *weak unlinkability*.

In the following theorem, we show that our rejection-sampling procedure makes the standard and adapted signature distributions identical. The full proof is deferred to Appendix A.2.

Theorem 1 (Rejection Sampling Algorithm). *Let $R_q = \mathbb{Z}_q[X]/(X^d + 1)$, fix $\gamma, \kappa \in \mathbb{N}$, and recall the distributions:*

$$\mathcal{D}_0 : \mathbf{x} \xleftarrow{\$} \mathcal{S}_{\gamma-\kappa}^m, \quad \mathcal{D}_1 : \mathbf{x} = \hat{\mathbf{z}} + \mathbf{y}, \text{ where } \hat{\mathbf{z}} \xleftarrow{\$} \mathcal{S}_{\gamma-\kappa-1}^m, \mathbf{y} \xleftarrow{\$} \mathcal{S}_1^m.$$

Let $g(\mathbf{x})$ and $f(\mathbf{x})$ be the probability mass functions of $\mathcal{D}_0$ and $\mathcal{D}_1$ respectively (both supported on $\mathcal{S}_{\gamma-\kappa}$). For a bounding constant $M \geq \sup_{\mathbf{x} \in \mathcal{S}_{\gamma-\kappa}} f(\mathbf{x})/g(\mathbf{x})$, consider the following algorithm A:

1. Sample $Y \xleftarrow{\$} \mathcal{D}_0$ (i.e., Y is uniform over $\mathcal{S}_{\gamma-\kappa}^m$).
2. Sample $U \xleftarrow{\$} \text{Uniform}[0, 1]$.
3. If $U \leq f(Y)/(Mg(Y))$, **output** $X = Y$; else reject and repeat.

Then the law of X (conditioned on acceptance) is exactly $\mathcal{D}_1$, and the acceptance probability of one iteration equals $1/M$.

Operational Efficiency of Theorem 1. Theorem 1 guarantees that, for any admissible bounding constant M , a single iteration accepts with probability $1/M$, whence the expected number of iterations satisfies $\mathbb{E}[N] = M$. It therefore suffices to instantiate M for our concrete distributions and parameters. With $n = \ell = 4$, $d = 256$, $\kappa = 60$ and $m = n + \ell$ as in Table 1, and the masking bound $\gamma = \kappa d(n + \ell)$, the mass ratio admits the bound

$$\sup_{\mathbf{x} \in \mathcal{S}_{\gamma-\kappa}^m} \frac{f(\mathbf{x})}{g(\mathbf{x})} < \exp\left(\frac{(n + \ell)d}{\gamma}\right) = \exp\left(\frac{1}{\kappa}\right) = \exp\left(\frac{1}{60}\right) = 1.01680\dots,$$

Setting $M := 1.0169$ thus satisfies the hypothesis of Theorem 1, which yields an expected number of iterations $\mathbb{E}[N] = 1.0169$ and an acceptance probability of $1/M \approx 98.34\%$.

Pre-Verify Soundness Refinement. To overcome the inherent obstacle in lattice-based relation statement verification, we introduce a structural refinement to the LAS⁺ protocol by shifting to the aforementioned NIZK-augmented relations. Specifically, the public statement is elevated to a composite $\mathbf{Y} = (Y, \pi)$. The PreSign and PreVerify algorithms are thereby refined to take $\mathbf{Y}$ as the statement for the NIZK-augmented relation $\mathfrak{R}_{\mathbf{A}}$ defined above. The NIZK can be instantiated using an exact proof system (e.g., Esgin et al. [5]) whose soundness ensures ($\|\mathbf{y}\|_{\infty} \leq 1$). Finally, Figure 1 illustrates the modified LAS⁺ scheme, with the NIZK parsing and verification steps highlighted.

4.2 Security Proofs

We prove that our scheme satisfies *extractability*, *unique extractability*, and *pre-verify soundness* as defined in [6], as well as our modified version of *weak unlinkability*. Due to space constraints, we include the proof of Theorem 2 in the Appendix A.2, while deferring the full proofs of the remaining theorems to the full version of our paper.

Theorem 2 (Extractability). *If M-SIS_{n,n+ℓ+1,q,β} is hard for $\beta = 2\gamma\sqrt{d(n + \ell)}$ and M-LWE_{ℓ,n,q} is hard, then the scheme LAS⁺ satisfies extractability in the random oracle model.*

Theorem 3 (Unique Extractability). *If M-SIS_{n,n+ℓ,q,β} is hard for $\beta = 2\gamma\sqrt{d(n + \ell)}$ and M-LWE_{ℓ,n,q} is hard, then the scheme LAS⁺ satisfies unique extractability in the random oracle model.*

Theorem 4 (Weak Unlinkability). *If $M\text{-LWE}_{\ell,n,q}$ is hard, the LAS^+ scheme satisfies weak unlinkability in the random oracle model, provided that the rejection-sampling procedure in Theorem 1 ensures statistical indistinguishability between standard and adapted signatures.*

Theorem 5 (Pre-Verify Soundness). *Let $\mathfrak{R}_A$ be the standard base relation defined by $A\mathbf{y} = Y \pmod{q}$ and $\|\mathbf{y}\|_\infty \leq 1$. If the underlying NIZK protocol satisfies standard soundness with respect to $\mathfrak{R}_A$, then the LAS^+ scheme achieves computational pre-verify soundness.*

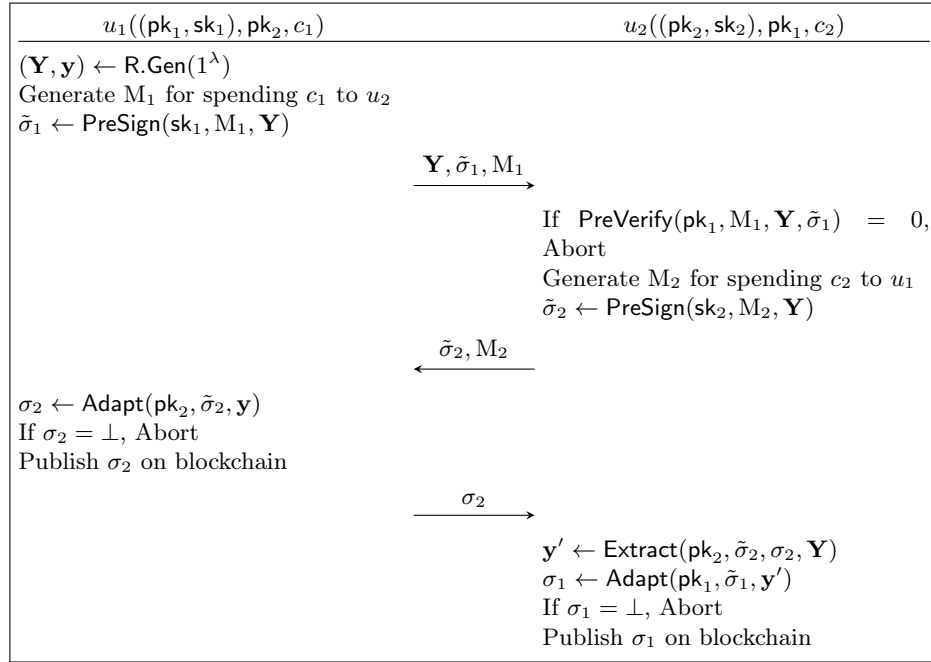


Fig. 2. Protocol diagram for atomic swap using LAS^+ .

5 Application: Unlinkable Atomic Swap

In this section, we focus on the application of our enhanced scheme LAS^+ to *atomic swaps*. An atomic swap using adaptor signatures [18] enables two users u_1, u_2 to exchange coins c_1, c_2 across (possibly different) UTXO-based blockchains by locking funds into transactions, whose spending conditions are tied to a pre-signature related to a public statement $\mathbf{Y}$.

The original LAS construction [4] requires manual, application-specific zero-knowledge proofs to ensure fairness. In contrast, LAS^+ provides a more general

template (Figure 2). It natively integrates an NIZK into its augmented relation to achieve *pre-verify soundness*. This eliminates the need for ad-hoc ZKP patches at the application layer. Parties simply generate an augmented statement-witness pair $(\mathbf{Y}, \mathbf{y})$ and exchange pre-signatures. They can then rely on **PreVerify** to detect maliciously injected statements. Consequently, LAS⁺ serves as a clean, general primitive for upper-layer protocols.

Furthermore, our analysis reveals a privacy gap in LAS: standard and adapted signatures have distinguishable distributions. This variance allows observers to link swap-related signatures on-chain. Instantiating the atomic swap with LAS⁺ resolves this issue and achieves weak unlinkability. Specifically, a rejection-sampling transformation ensures that **Sign** and **Adapt** produce identically distributed signatures. Crucially, external adversaries only see the final signatures published on the blockchain; they cannot observe the private off-chain pre-signatures. Given this restricted view, this distributional equivalence directly guarantees on-chain privacy. Thus, LAS⁺ yields an atomic swap that achieves post-quantum security, algorithmic fairness (via *pre-verify soundness*), and on-chain privacy (via *weak unlinkability*) without complex application-level interventions.

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A Appendix

A.1 Security Assumptions: Module-SIS and Module-LWE

Following [15], we work with the Hermite Normal Form (HNF) representation of the M-SIS matrix, which is known to be as hard as using a uniformly random matrix.

Definition 10 (M-SIS _{n,m,q,β_{SIS}} [4, 9]). *The M-SIS problem with parameters $m > n > 0$ and $0 < \beta_{\text{SIS}} < q$ is defined as follows. Sample $\mathbf{A}' \xleftarrow{\$} R_q^{n \times (m-n)}$ and set $\mathbf{A} = [\mathbf{I}_n \parallel \mathbf{A}'] \in R_q^{n \times m}$. Given $\mathbf{A}$, the goal is to find a non-zero short vector $\mathbf{v} \in R_q^m$ such that*

$$\mathbf{A}\mathbf{v} = 0 \bmod q \quad \text{and} \quad \|\mathbf{v}\|_{\infty} \leq \beta_{\text{SIS}}.$$

For M-LWE we use the standard small-secret variant, where both the secret and the error are chosen from $\{-1, 0, 1\}$; this setting is widely adopted in practice, e.g., in Dilithium [3].

Definition 11 (M-LWE _{ℓ,m,q} [4, 9]). *The M-LWE problem with parameters (ℓ, m, q) is defined as distinguishing between the following two distributions:*

- (i) $(\mathbf{A}, \mathbf{b}) \quad \xleftarrow{\$} R_q^{m \times \ell} \times R_q^m,$
- (ii) $(\mathbf{A}, \mathbf{A}\mathbf{s} + \mathbf{e}) \quad \text{where } \mathbf{A} \xleftarrow{\$} R_q^{m \times \ell}, \mathbf{s} \xleftarrow{\$} \mathcal{S}_1^{\ell}, \mathbf{e} \xleftarrow{\$} \mathcal{S}_1^m.$

It is well known that sampling $\mathbf{s}$ and $\mathbf{e}$ from a wider range $\mathcal{S}_{\gamma}$ with $\gamma > 1$ only makes the problem harder, i.e., yields a weaker assumption. In particular, under the M-LWE _{ℓ,m,q} assumption, the term $\mathbf{t} = \mathbf{A}\mathbf{s} + \mathbf{e}$ is computationally indistinguishable from a uniformly random vector in R_q^m , even when $\mathbf{s} \xleftarrow{\$} \mathcal{S}_{\gamma}^{\ell}$ and $\mathbf{e} \xleftarrow{\$} \mathcal{S}_{\gamma}^m$ for any $\gamma > 1$.

A.2 Proof of Theorems

Theorem 1 (Rejection Sampling Algorithm) Let $R_q = \mathbb{Z}_q[X]/(X^d + 1)$, fix $\gamma, \kappa \in \mathbb{N}$, and recall the distributions:

$$\mathcal{D}_0 : \mathbf{x} \xleftarrow{\$} \mathcal{S}_{\gamma-\kappa}^m, \quad \mathcal{D}_1 : \mathbf{x} = \hat{\mathbf{z}} + \mathbf{y}, \text{ where } \hat{\mathbf{z}} \xleftarrow{\$} \mathcal{S}_{\gamma-\kappa-1}^m, \mathbf{y} \xleftarrow{\$} \mathcal{S}_1^m.$$

Let $g(\mathbf{x})$ and $f(\mathbf{x})$ be the probability mass functions of $\mathcal{D}_0$ and $\mathcal{D}_1$ respectively (both supported on $\mathcal{S}_{\gamma-\kappa}$). For a bounding constant $M \geq \sup_{\mathbf{x} \in \mathcal{S}_{\gamma-\kappa}} f(\mathbf{x})/g(\mathbf{x})$, consider the following algorithm A:

1. Sample $Y \xleftarrow{\$} \mathcal{D}_0$ (i.e., Y is uniform over $\mathcal{S}_{\gamma-\kappa}^m$).
2. Sample $U \xleftarrow{\$} \text{Uniform}[0, 1]$.
3. If $U \leq f(Y)/(Mg(Y))$, **output** $X = Y$; else reject and repeat.

Then the law of X (conditioned on acceptance) is *exactly* $\mathcal{D}_1$, and the acceptance probability of one iteration equals $1/M$.

Proof. Since the proposal distribution g and the target distribution f share the same finite support $\Omega = \mathcal{S}_{\gamma-\kappa}$, we analyze the cumulative distribution of the output X . For any threshold x , conditional probability yields

$$\begin{aligned} \Pr[X \leq x] &= \Pr[Y \leq x \mid \text{accept}] \\ &= \frac{\Pr[Y \leq x \wedge \text{accept}]}{\Pr[\text{accept}]}. \end{aligned}$$

Let the acceptance event be $\mathbf{A} = \{U \leq \frac{f(Y)}{Mg(Y)}\}$. Given that $U \sim \text{Uniform}[0, 1]$ is independent of Y , we evaluate the numerator via the law of total probability:

$$\begin{aligned} \Pr[Y \leq x \wedge \mathbf{A}] &= \sum_{y \leq x} \Pr\left[U \leq \frac{f(y)}{Mg(y)} \mid Y = y\right] \Pr[Y = y] \\ &= \sum_{y \leq x} \frac{f(y)}{Mg(y)} g(y) \\ &= \frac{1}{M} \sum_{y \leq x} f(y). \end{aligned}$$

Similarly, since f is a valid probability mass function normalized over Ω , the overall acceptance probability is precisely

$$\begin{aligned} \Pr[\mathbf{A}] &= \sum_{y \in \Omega} \frac{f(y)}{Mg(y)} g(y) \\ &= \frac{1}{M} \sum_{y \in \Omega} f(y) = \frac{1}{M}. \end{aligned}$$

Substituting these results back into the initial expression gives $\Pr[X \leq x] = \sum_{y \leq x} f(y)$. Therefore, the CDF of X exactly matches that of the target distribution f , implying $X \sim \mathcal{D}_1$, and the algorithm succeeds with probability exactly $1/M$.

Theorem 2 (Extractability) If $\text{M-SIS}_{n,n+\ell+1,q,\beta}$ is hard for $\beta = 2\gamma\sqrt{d(n+\ell)}$ and $\text{M-LWE}_{\ell,n,q}$ is hard, then the scheme LAS^+ satisfies *extractability* in the random oracle model.

Proof. Let $\mathcal{F}$ be a PPT adversary who wins the Extractability game with a non-negligible probability. We construct an adversary $\mathcal{S}$ that utilizes $\mathcal{F}$ to solve the M-SIS $_{n,n+\ell+1,q,\beta}$ problem.

Let $\beta = 2\gamma\sqrt{d(n+\ell)}$. $\mathcal{S}$ receives an M-SIS challenge matrix $\mathbf{B} = [\mathbf{I}_n \parallel \mathbf{A}' \parallel \mathbf{a}] \in R_q^{n \times (n+\ell+1)}$, where $\mathbf{A}' \xleftarrow{\$} R_q^{n \times \ell}$ and $\mathbf{a} \xleftarrow{\$} R_q^n$. Let $\mathbf{A} = [\mathbf{I}_n \parallel \mathbf{A}']$.

Setup. $\mathcal{S}$ establishes $\mathbf{A}$ and a random oracle H as the public parameters. To generate the public key, $\mathcal{S}$ samples $\mathbf{r}' \xleftarrow{\$} \mathcal{S}_1^{n+\ell}$, sets $\mathbf{r} = \begin{pmatrix} \mathbf{r}' \\ 1 \end{pmatrix}$, and computes $\text{pk} = \mathbf{B}\mathbf{r} = \mathbf{A}\mathbf{r}' + \mathbf{a}$. $\mathcal{S}$ then sends pk to $\mathcal{F}$. By the M-LWE $_{\ell,n,q}$ assumption, $\mathbf{A}\mathbf{r}' + \mathbf{a}$ is computationally indistinguishable from a uniformly random vector, ensuring pk is computationally indistinguishable from an honest public key.

Oracle Simulation.

- $\mathcal{O}_{\text{Sign}}(m)$: $\mathcal{S}$ samples $\mathbf{z} \xleftarrow{\$} \mathcal{S}_{\gamma-\kappa}^{n+\ell}$ and $c \xleftarrow{\$} \mathcal{C}$, then programs the random oracle as $c = H(\text{pk}, \mathbf{A}\mathbf{z} - c \cdot \text{pk}, m)$. If a collision occurs in H , $\mathcal{S}$ aborts. Otherwise, it returns $\sigma = (c, \mathbf{z})$.
- $\mathcal{O}_{\text{PreSign}}(m, \mathbf{Y})$: $\mathcal{S}$ parses the augmented statement as $\mathbf{Y} = (Y, \pi)$. It samples $\hat{\mathbf{z}} \xleftarrow{\$} \mathcal{S}_{\gamma-\kappa-1}^{n+\ell}$ and $c \xleftarrow{\$} \mathcal{C}$, programming the oracle as $c = H(\text{pk}, \mathbf{A}\hat{\mathbf{z}} - c \cdot \text{pk} + Y, m)$. It aborts on collision; otherwise, returns $\tilde{\sigma} = (c, \hat{\mathbf{z}})$.
- $\mathcal{O}_{\text{NewY}}$: For the i -th query, $\mathcal{S}$ samples a short vector $\mathbf{r}_i \xleftarrow{\$} \mathcal{S}_1^{n+\ell}$ and computes $Y_i = \mathbf{A}\mathbf{r}_i - \mathbf{a}$. Relying on the zero-knowledge property of the underlying NIZK, $\mathcal{S}$ invokes the NIZK simulator to generate a valid-looking proof π_i for Y_i without using a valid witness. It records $(i, \mathbf{r}_i, Y_i, \pi_i)$ and returns the augmented statement $\mathbf{Y}_i = (Y_i, \pi_i)$. Crucially, since the short randomness $\mathbf{r}_i$ is sampled internally by $\mathcal{S}$ and remains completely hidden from the adversary $\mathcal{F}$, the term $\mathbf{A}\mathbf{r}_i$ is computationally indistinguishable from a uniformly random vector in R_q^n under the M-LWE $_{\ell,n,q}$ assumption. Consequently, the shifted public statement $Y_i = \mathbf{A}\mathbf{r}_i - \mathbf{a}$ also appears uniformly random to $\mathcal{F}$, ensuring it is distributed identically to an honest relation generation.

Forgery & Analysis. Eventually, $\mathcal{F}$ outputs a target message m^* and a forged signature $\sigma^* = (c, \mathbf{z})$. To analyze this forgery, we distinguish three mutually exclusive cases regarding the challenge c :

Case 1: PreSign queried, but not NewY. The forgery relies on a pre-signature $\tilde{\sigma} = (c^*, \hat{\mathbf{z}})$ obtained from $\mathcal{O}_{\text{PreSign}}(m^*, \mathbf{Y})$, such that $c = c^*$, but $\mathbf{Y} = (Y, \pi)$ was maliciously crafted by $\mathcal{F}$. From the verification equations, we have:

$$c = H(\text{pk}, \mathbf{A}\mathbf{z} - c \cdot \text{pk}, m^*) \quad \text{and} \quad c^* = H(\text{pk}, \mathbf{A}\hat{\mathbf{z}} - c^* \cdot \text{pk} + Y, m^*).$$

Since $c = c^*$, it strictly implies $\mathbf{A}\mathbf{z} - c \cdot \text{pk} = \mathbf{A}\hat{\mathbf{z}} - c^* \cdot \text{pk} + Y$, which simplifies to $\mathbf{A}(\mathbf{z} - \hat{\mathbf{z}}) = Y$. Let $\mathbf{s} = \mathbf{z} - \hat{\mathbf{z}}$. Given $\|\mathbf{z}\|_\infty \leq \gamma - \kappa$ and $\|\hat{\mathbf{z}}\|_\infty \leq \gamma - \kappa - 1$, we obtain $\|\mathbf{s}\|_\infty \leq 2(\gamma - \kappa)$. Thus, the extracted witness $\mathbf{s}$ satisfies $(\mathbf{Y}, \mathbf{s}) \in \hat{\mathcal{R}}'_\mathbf{A}$. However, the winning condition of the Extractability game explicitly requires that the extracted witness must *not* be valid (i.e., $(\mathbf{Y}, \text{Extract}(\text{pk}, \tilde{\sigma}, \sigma^*, \mathbf{Y})) \notin \hat{\mathcal{R}}'_\mathbf{A}$). Therefore, this scenario strictly contradicts the condition of a successful forgery and $\mathcal{S}$ aborts.

Case 2: Both PreSign and NewY queried. The forgery matches a pre-signature generated under an augmented relation $\mathbf{Y}$ previously returned by $\mathcal{O}_{\text{NewY}}$. Recall that $\mathcal{S}$ generated its base statement as $Y = \mathbf{A}\mathbf{r}_i - \mathbf{a}$. Following the algebraic derivation in Case 1, we again have $\mathbf{A}(\mathbf{z} - \hat{\mathbf{z}}) = Y$. Substituting Y , we get:

$$\mathbf{A}(\mathbf{z} - \hat{\mathbf{z}}) = \mathbf{A}\mathbf{r}_i - \mathbf{a} \implies \mathbf{A}(\mathbf{z} - \hat{\mathbf{z}} - \mathbf{r}_i) = -\mathbf{a}.$$

Let $\mathbf{s} = \mathbf{z} - \hat{\mathbf{z}} - \mathbf{r}_i$. This equation is algebraically equivalent to $\mathbf{B} \begin{pmatrix} \mathbf{s} \\ 1 \end{pmatrix} = \mathbf{0}$. The infinity norm is bounded by $\|\mathbf{s}\|_\infty \leq \|\mathbf{z}\|_\infty + \|\hat{\mathbf{z}}\|_\infty + \|\mathbf{r}_i\|_\infty \leq 2(\gamma - \kappa)$. Converting to the L_2 norm, we have $\left\| \begin{pmatrix} \mathbf{s} \\ 1 \end{pmatrix} \right\| \leq \beta$. Thus, $\mathcal{S}$ successfully finds a valid short solution to the M-SIS challenge.

Case 3: No valid pre-signature used. For all queried signatures $\sigma = (c^*, \mathbf{z}^*)$, we have $c^* \neq c$. Thus, the oracle never programmed c for m^* . By the Forking Lemma, $\mathcal{S}$ can rewind $\mathcal{F}$ to the exact point of the random oracle query to obtain a second valid forgery $\sigma' = (c', \mathbf{z}')$ on m^* , such that $c' \neq c$ but the commitment $\mathbf{w}$ remains identical:

$$\mathbf{A}\mathbf{z}' - c' \cdot \text{pk} = \mathbf{A}\mathbf{z} - c \cdot \text{pk} \implies \mathbf{A}(\mathbf{z}' - \mathbf{z}) = (c' - c)\text{pk}.$$

Recall from Setup that $\text{pk} = \mathbf{B}\mathbf{r}$, and $\mathbf{A}\mathbf{x} = \mathbf{B} \begin{pmatrix} \mathbf{x} \\ 0 \end{pmatrix}$ for any vector $\mathbf{x}$. We can rewrite the above as:

$$\mathbf{B} \begin{pmatrix} \mathbf{z}' - \mathbf{z} \\ 0 \end{pmatrix} = \mathbf{B}((c' - c)\mathbf{r}) \implies \mathbf{B} \left[(c' - c) \begin{pmatrix} \mathbf{r}' \\ 1 \end{pmatrix} - \begin{pmatrix} \mathbf{z}' - \mathbf{z} \\ 0 \end{pmatrix} \right] = \mathbf{0}.$$

Since the last component of this solution vector is $(c' - c) \cdot 1 - 0 = c' - c \neq 0$, the vector is non-zero. The norm of this solution is strictly bounded by $2\gamma\sqrt{d(n+\ell)} = \beta$ (since $\gamma \gg \kappa$). Consequently, $\mathcal{S}$ extracts a non-trivial short solution to the M-SIS problem, concluding the proof.